

State-correlated reaction dynamics unveiled in full from a single product-image measurement

Corresponding Author: Dr Huilin Pan

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors investigate the reaction dynamics of $F + CH_4$ in a joint experimental and theoretical study. They use crossed beam velocity map imaging in combination with one photon ionization by a VUV photon from a free electron laser. This combination allows them to record product state specific differential cross sections which they term correlated differential cross sections. They complement the experimental results by high level quantum dynamics simulations. They find a remarkable agreement between experiment and theory. The study is well executed and the experimental and theoretical data is convincing and support the arguments made by the authors (once one understands them) but the presentation in the manuscript is not at the same level as the data or its analysis. The study might be suitable for publication in Nature Communications after major revisions of the data presentation and the manuscript.

There is little to no discussion on prior studies, experimental or theoretical, on the reaction except for the experimental one done by the authors before. They should take more care about this and I find this not up to common standards.

The reaction has been studied by part of the authors using the same (according to extended data Fig 1) except for the ionization scheme used for the CH_3 radical. In earlier studies REMPI was used and each state was investigated in its own experiment whereas now different product states can be monitored in the same experimental run, respectively image. The authors highlight the new experimental capabilities as a major breakthrough but give surprisingly little experimental details.

I find the language used throughout the manuscript unnecessarily fanciful and little to forward for me. A simpler language would help with understanding because I was often confused reading the manuscript. This is a starting with the abstract. What is the conventional perception the authors allude to throughout the manuscript? The use of "all-important" is not followed by an explanation as to why? The authors give bold statements and arguments but then these are very unspecific or lack facts and numbers. They state on page 5 that the present experiment has a "significantly higher resolution" but there are no numbers given: neither on the reactant beam uncertainties, the collision energy or the product velocity spread or the specs of the VUV beam. Only the temperature of the CH_4 is given as 8K in conjunction with the angular momentum discussion.

The discussion on the importance or not importance of a prior found dynamical resonance to the presented data interpretation is so convoluted that it almost seems like dropping a key word. It requires the caption of Extended Data Fig. 3 and the SI to understand. The manuscript should stand by itself. The whole paragraph somehow reads backwards. I need several attempts to make some sense of it. And I am still not sure if I got right in the end: The dynamical does not play a significant role in for the present study, right? If my interpretation of the paragraph is correct, why put so

Figure 1

I consider the background subtracted image not really a difference image. Part of the authors referred to experiments with and without vibrationally excited reactant to difference images. This is a real difference image. Background subtraction needs to be done and properly explained (as done here) but it is no difference image.

I am missing a legend in figure 1A and 1B for the scaling for the color schemes used in the histograms. Is it linear or logarithmic, absolute or normalized? Why use the same line shape and color for two different kinematic cut-offs? Please make them distinguishable.

Page 5, first full paragraph

The first sentence is really confusing and long. What is a “recoiled velocity”?

Figure 7

The width of the used fitting functions and the significance of the relative contributions cannot be validated because no energy spread is given. Why use km/s as unit?

Page 8, Line 2

Can the authors please indicate why part of the image they are evaluating? In figure 1 only 0° and 180° are indicated. For experts is obvious but not for a general chemistry audience.

Page 8, Line 15

The authors refer to “proper scaling” what is scaled how and why? More information is needed.

Figure 3A

What is the unit for the “correlated ICS” given that the data comes from a crossed beam experiment and the associated velocity map image. The data is not really an integral cross section but rather a relative branching ratio of the different product channels and should be labeled as such. Is there an error associated to it?

Page 11 Line 22

I do not understand the sentence. Something is missing. The authors should check if “auto-correction” did some unintended corrections.

Page 12, Line 9

“does not directly invoke in bond cleavage and formation”

Please check wording.

Page 12, Line 15+16

I do not understand the meaning of this sentence in relation to reaction centers, respectively dynamics. The authors might want to check the language.

Page 12, Line 26

Shouldn't that be forward scattered CH₃ when the HF is scattered backward. Also see next sentence. I guess something got mixed up.

Reviewer #2

(Remarks to the Author)

The authors (and others) have previously carried out extensive studies on the dynamics of the reaction $F + CH_4 \rightarrow CH_3 + HF$ and deuterated analogs, in which they have probed CH₃ products in their vibrational ground state and in various umbrella-mode (ν_2) vibrationally excited states ($\nu_2 = 0, 1, 2, 3$) via REMPI and velocity-map imaging. These studies have revealed the pair-correlated scattering distributions of the CH₃ with various vibrational states of the HF (or HD) coproduct. The present study extends this work by using VUV photoionization at 118 nm as a universal probe to enable simultaneous detection of all rovibrational states of CH₃ formed in the chemical reaction. The new experimental results are interpreted with the help of comprehensive six-dimensional quantum dynamics simulations.

In addition to the product channels previously characterised, the paper identifies, and importantly, quantifies contributions from asymmetric-stretch (ν_1) excited CH₃ products. These have been observed before using REMPI detection, but one-photon ionisation makes it much more straightforward to determine the relative contributions from different product channels. The authors present product state distributions and pair-correlated angular distributions in excellent agreement with the theoretical predictions, and provide a detailed commentary on the mechanisms leading to the observed angular scattering.

This is a comprehensive, generally well-written, and very high quality piece of work, which fills in some additional dynamical details for the benchmark $F + CH_4$ reaction. However, I was extremely surprised at the number of times the authors stated the novelty of their 118 nm universal ionization approach, given that this ionization method, as well as universal ionization using other VUV wavelengths, has been used in velocity-map imaging studies in a number of different laboratories for well over a decade. A few examples of which I am immediately aware include Fan et al, J. Chem. Phys., 123, 204301 (2005), Ashfold et al, J. Phys. Chem. Lett, 8, 3440 (2017), Gardiner et al, Phys. Chem. Chem. Phys., 17, 4096 (2015), Lipciuc et al, J. Chem. Phys., 147, 013941 (2017), and various other papers from these and other groups, but I am sure there are numerous others. None of this previous work was mentioned or referenced in the manuscript. In my view, this is the one major shortcoming of the paper as it stands. There are also a few minor typos, etc, that the authors may want to correct in a revised version of the manuscript.

1. Abstract: remove the initial ‘The’. Also, “electron-impact ionization” is now usually referred to simply as “electron ionization”.

2. Page 3, line 13: "a state-resolved detection" should just be "state-resolved detection".
3. Page 11, line 8: lines 8-11 could do with a small amount of rewording, for clarity, e.g. "Thereby, the discrepancies in the forward peaks, the imprint of the resonance pathway, would be reduced if the reactants $\text{CH}_4(j = 1,2)$ were also included in the calculation. More significantly, the overall distribution evolves from being predominantly forward peaking for $(0_0,3)$ to exhibiting marked sideways scattering for $(2_2,2)$, and then to essentially backward-scattering for $(0_0,2)$."
4. Page 11, line 16: "recoiled-energy" should be "recoil energy"
5. Page 12, line 9: I am not sure what the authors mean by "The CH_3 -moiety does not directly invoke in bond cleavage and formation". Do they mean that the CH_3 is not directly *involved* in bond cleavage and formation?
6. Page 12, line 11: should be "...products attests to the strong coupling..."
7. Page 12, line 15: I am not sure what the authors mean when they say that the transferred H atom "is always sprouting away" from the centre of mass of CH_4 . Are they just referring to the chemical structure of CH_4 ? If so, it might be clearer to say that the breaking C-H bond is always directed away from the centre of mass of CH_4 , which lies on the C atom. However, I find it difficult to think of an example in which a breaking bond would not be pointing away from the centre of mass of the molecule within which it resides, so perhaps the authors could find a clearer way of making their point in this regard, perhaps with the aid of a diagram?

Reviewer #3

(Remarks to the Author)

The VUV single photon ionization method has been well known as a soft and universal product detection scheme, while universal detection has been linked with no product state specificity all the time previously. In this study, Pan et al. demonstrated an experiment combining the universality of VUV photoionization for the reaction product detection and the high velocity resolution achievable in a three-dimensional VMI which can handle relatively large reaction or ionization volumes of the reaction products, and they achieved universality and state-specificity for product detection simultaneously. They successfully mapped out almost all the important product pair-correlated dynamical attributes of the prototypical polyatomic reaction $\text{F} + \text{CH}_4 \rightarrow \text{CH}_3(\text{vi}) + \text{HF}(\text{v})$ from a single product image. Six-dimensionality quantum dynamics calculations are performed and found to be able to reproduce nearly all the aspects observed in the experiment, and have provided new insights into this important prototypical polyatomic reaction. Such a method is a very useful addition to the community of reaction dynamics, and can be definitely used in the studies of many other reactions, I thus recommend its publication in Nature Communications. The whole paper is very nicely presented, and I only have two comments for the authors to consider.

1. The universal VUV single photoionization detection is an important addition to the product state-specific detection method, and can provide most of the time qualitative information for the full dynamical picture. However, its ability may not be overstated. In the study of the $\text{F} + \text{CH}_4 \rightarrow \text{CH}_3(\text{vi}) + \text{HF}(\text{v})$, one of the reasons why it is very successful is that the products are all rotationally cold, which enables the resolve of the pair-correlated vibrational distributions. For reaction products with significant rotational excitation, the complete resolve will become difficult. At the same time, previous state-specific studies also play an important role in analyzing the data in this study. I thus believe that the proper combination of the universal and state-specific methods would become a powerful method in studying reaction dynamics in the future, as demonstrated in this study.
2. For the last paragraph on page 12, a classical physical picture is described to explain the "observed correlation that low umbrella-excited CH_3 prefers backward scattering, whereas higher umbrella-excited CH_3 yields more in the sideway and forward directions." Two points that may need be explained a little more: first, it seems that both the large and small impact collisions involve the flatten motion of the CH_3 moiety, while one results in the umbrella vibrational excitation, and the other dose not? Second, the above picture only applies to processes within the same HF vibrational excitation, for example, for $\text{HF}=3$, the products are mainly forward scattered, even though the umbrella motion of CH_3 is not excited. Is it possible to make it clearer why this picture strongly depends on the excitation of the co-product HF?

Reviewer #4

(Remarks to the Author)

The authors describe, implement, and test a VMI experiment capable of what they call "universal" state-resolved product selection. The well-studied $\text{F} + \text{CH}_4$ system is considered, and the experiment is shown to resolve product states of CH_3 alongside HF. The instrument is validated against quantum dynamics, which on its own is a substantial achievement of the work.

The demonstration given here is notable and, in my opinion, is appropriate for publication in a high-impact journal. Unfortunately, some awkward English and rushed descriptions make this short paper difficult to understand in places. I would specifically suggest improving the English in the abstract and elaborating on key discussion points. As written, the reader needs to already be familiar with VMI studies of this system to understand the paper.

For example, the notation of the observed state pairs, e.g., (23, 2), are not explicitly explained and instead must be inferred.

I would suggest clarifying what aspects of the new design are responsible for the breakthrough here. The earlier work did not get information for all the vibrational levels of CH₃ at once, like here, but I wasn't sure why. Likewise, is there additional information in this experiment, as compared to earlier "3D" VMI studies on this system, or is the advance here that the information is gathered more directly?

Finally, I wondered if plots like Fig. 1A could be produced from the calculations?

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors answered my questions and cleared up some confusion. The changes to the text make the manuscript easier to understand. The manuscript is suitable for Nature Communications. Maybe, the authors want to consider some minor comments:

In the abstract and on page 5 they refer to "previously inaccessible insights". Why not state at this point what was previously inaccessible more plainly. The wording works in the abstract but on page five it reads strange with the kind of left confused at as to what is new- The answer is only given later on page 6.

I now understand the author's use of forward and backward in the relative frame of reference. Maybe readers would benefit from an addition of the relative orientation of the incident beams with respect to the scattering angle as indicated in figure 1.

I find "all-important" a presumptuous phrase to use. Why use it at all? For me, it adds nothing to the paper but confusion because it has no direct scientific meaning.

A comment:

In the authors refer to space and length of the communication with respect to the limited number of references to earlier studies. Length might be an argument, number of allowed references is not. The author guidelines state that up to 70 references are allowed or considered reasonable. The current number of references is 44 (three more than in the original submission). I still think this is a major point to consider. The authors do not need to write a "review" style introduction but the references to prior studies on either the reaction studies or the methodology are limited.

Reviewer #2

(Remarks to the Author)

The authors have responded comprehensively to the suggestions of the referees, and amended their manuscript accordingly. Assuming the other referees are in agreement, I believe that the paper is now suitable for publication in Nature Communications.

Reviewer #3

(Remarks to the Author)

The authors have addressed all my comments in proper ways, I thus recommend its publication in Nature Communications in its current form.

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Author responses to the reviewers' reports on NCOMMS-25-55690-T

Reviewer #1 (Remarks to the Author):

The authors investigate the reaction dynamics of $F + CH_4$ in a joint experimental and theoretical study. They use crossed beam velocity map imaging in combination with one photon ionization by a VUV photon from a free electron laser. This combination allows them to record product state specific differential cross sections which they term correlated differential cross sections. They complement the experimental results by high level quantum dynamics simulations. They find a remarkable agreement between experiment and theory. The study is well executed and the experimental and theoretical data is convincing and support the arguments made by the authors (once one understands them) but the presentation in the manuscript is not at the same level as the data or its analysis. The study might be suitable for publication in Nature Communications after major revisions of the data presentation and the manuscript.

[Reply]: Thank you for the positive comments about our works. We will try to improve our presentation in the revision to match up with the data and analysis. By the way, there may be a misunderstanding that the VUV photon used in this work was generated by a third-harmonics process, not from a free electron laser.

There is little to no discussion on prior studies, experimental or theoretical, on the reaction except for the experimental one done by the authors before. They should take more care about this and I find this not up to common standards.

[Reply]: Yes, there is a very long list of experimental and theoretical literatures on this benchmark reaction over the past decades because of both fundamental and technological (e.g., chemical laser) relevance. It will take at least a long paragraph just to glance over them without detailed descriptions. Thus, in this short communication only the most relevant aspects (product state-resolved angular distributions) to the present work are discussed. Discussion on many of those earlier literatures can be found in the cited publications of this work. A sentence has been added on page 5, line 6 to call attention of readers.

The reaction has been studied by part of the authors using the same (according to extended data Fig 1) except for the ionization scheme used for the CH_3 radical. In earlier studies REMPI was used and each state was investigated in its own experiment whereas now different product states can be monitored in the same experimental run, respectively image. The authors highlight the new experimental capabilities as a major breakthrough but give surprisingly little experimental details.

[Reply]: That is not really a new capability, but rather a realization of the potential (or the feature) of our experimental setup and an execution to its current limit. This key feature (capable of achieving space-focusing effect at the same time of 2D VMI) was explicitly elaborated. To be clear, we rewrite that part, starting from page 4, line 29 to page 5, line 4. In retrospect, the potential of the present approach for achieving both universal and state-specific detection of crossed-beam scattering products is so obvious that we were surprising that no one, including ourselves, has made serious attempts over past years.

I find the language used throughout the manuscript unnecessarily fanciful and little to forward for me. A simpler language would help with understanding because I was often confused reading the manuscript. This is a starting with the abstract. What is the conventional perception the authors allude to throughout the manuscript? The use of "all-important" is not followed by an explanation as to why? The authors give bold statements and arguments but then these are very unspecific or lack facts and

numbers. They state on page 5 that the present experiment has a “significantly higher resolution” but there are no numbers given: neither on the reactant beam uncertainties, the collision energy or the product velocity spread or the specs of the VUV beam. Only the temperature of the CH₄ is given as 8K in conjunction with the angular momentum discussion.

[Reply]: Thanks for the comments. The abstract is rewritten in the revision. The ‘conventional perception’ referred to the common belief in the community that the universal detection is usually linked with no product state specificity. To be clear, we now replace the wording of ‘the conventional perception’ by a short sentence in plain language in the abstract.

We thought that “all-important” is apparent in the present context: In a typical chemical dynamics study, all important attributes are manifested in the product state- and angle-resolved distributions, which is revealed simultaneously by a single measurement in this study.

As for “significantly higher resolution”, the data speaks for itself: The structural ring-like features of umbrella-mode excited CH₃ products can clearly be identified as presented in Figs. 1 and 2. Those features are the convoluted results of all those instrument factors mentioned. And this observation is unprecedented, testifying higher resolution. Nevertheless, we now rewrite the paragraph (page 5, lines 7-13), and that wording is deleted.

The discussion on the importance or not importance of a prior found dynamical resonance to the presented data interpretation is so convoluted that it almost seems like dropping a key word. It requires the caption of Extended Data Fig. 3 and the SI to understand. The manuscript should stand by itself. The whole paragraph somehow reads backwards. I need several attempts to make some sense of it. And I am still not sure if I got right in the end: The dynamical does not play a significant role in for the present study, right? If my interpretation of the paragraph is correct, why put so

[Reply]: Yes, we believe that dynamical resonance does not play a significant role at the present collisional energy, which has explicitly been mentioned on page 11, lines 22-23: “At $E_c = 2.4$ kcal mol⁻¹ the direct abstraction mechanism prevails with minor contribution from the resonance pathway.”

The ‘signature’ of resonance in this reaction is mainly manifested in the HF($v = 3$) attributes, i.e., its branching fraction and forward peak. However, as discussed in the text (page 11, line 6), a classical mechanism could also contribute to those attributes. Unfortunately, it is difficult to quantify the relative contributions of two mechanisms (classical time-delay and quantum resonance). In addition, as shown in Extended Data Fig. 3, the QM results indicate that the initial CH₄ rotational excitation significantly reduces the HF($v = 3$) branching fraction, which we interpret as evidence that resonance is suppressed for the $j=1$ and 2 states. Considering that the experimental rotational temperature is 8 K ($n_j=0 : n_j=1 : n_j=2 = 0.31 : 0.54 : 0.15$), dynamical resonance should not play a significant role at the present collisional energy. Please note that although the theoretical forward scattering appears sharper than the experimental observations, the forward peak typically yields smaller contribution to total flux because of the solid angle factor. As such, a very sharp forward peak could only amount to small reactive flux.

Figure 1

I consider the background subtracted image not really a difference image. Part of the authors referred to experiments with and without vibrationally excited reactant to difference images. This is a real difference image. Background subtraction needs to be done and properly explained (as done here) but it is not a difference image.

[Reply]: The point is well taken. The word of ‘background subtracted image’ is used in the revision.

I am missing a legend in figure 1A and 1B for the scaling for the color schemes used in the histograms. Is it linear or logarithmic, absolute or normalized? Why use the same line shape and color for two different kinematic cut-offs? Please make them distinguishable.

[Reply]: The scaling for the color schemes is now added to Figs. 1A and 1B, as well as Extended Data Fig. 2.

We are not sure what you refer to; two red circles in Fig. 1A? If so, we now use two different colors.

Page 5, first full paragraph

The first sentence is really confusing and long. What is a “recoiled velocity”?

[Reply]: We now make the paragraph more concise, and hopefully it is more readable. The wording is now changed to “3D velocity information”.

Figure 7

The width of the used fitting functions and the significance of the relative contributions cannot be validated because no energy spread is given. Why use km/s as unit?

[Reply]: There is no Figure 7. We assume that you mean page 7, Figure 2 (?).

As explained in the text (page 8, lines 6-12), we used an independent measurement of λ -scanning the Q-head of $\text{CH}_3(0_0^0)$ REMPI band (the pink-shaded one in Fig. 2A), which included all apparatus functions, as a guide for the fitting functions to dissect the vibrational contributions. We believe that is best way one can analyze such a congested distribution. To be clear, we reword the sentences (page 8, lines 14-20).

Our measurement is in the speed domain, so the data is best analyzed in terms of speed, not in energy.

Page 8, Line 2

Can the authors please indicate why part of the image they are evaluating? In figure 1 only 0° and 180° are indicated. For experts is obvious but not for a general chemistry audience.

[Reply]: As stated in the text (page 8, line 5), the ring-like features in that angular range (see Fig 1A) are more apparent, thus more informative as to the vibrationally excited states. It then makes sense to start with that range. Once the vibrational shapes in that portion are determined, the full angular information can be obtained by scaling (up or down) each individual vibrational component, while keeping the shape intact, to fit the observation. For clarity, we rephrase the sentences (page 8, lines 14-20).

Page 8, Line 15

The authors refer to “proper scaling” what is scaled how and why? More information is needed.

[Reply]: See the above reply.

Figure 3A

What is the unit for the “correlated ICS” given that the data comes from a crossed beam experiment and the associated velocity map image. The data is not really an integral cross section but rather a relative branching ratio of the different product channels and should be labeled as such. Is there an error associated to it?

[Reply]: Agreed. The experimental figure axis is now relabeled as “vibrational branching fraction”. The actual error is difficult to quantify. It mostly comes from how well the fitting is to the observed distribution. Judging from what presented in Fig. 2, the room for further adjustments seems limited. So, Fig. 3 represents our “best” estimate for comparing with theoretical results.

Page 11 Line 22

I do not understand the sentence. Something is missing. The authors should check if “auto-correction” did some unintended corrections.

[Reply]: We are not sure about what this comment specifically referred to. Nonetheless, we rephrase the sentence (page 11, line 26); hopefully, it will clarify your concerns.

Page 12, Line 9

“does not directly invoke in bond cleavage and formation”

Please check wording.

[Reply]: We rephrase the sentence as “The CH₃-moiety is not directly involved in bond cleavage and formation.”

Page 12, Line 15+16

I do not understand the meaning of this sentence in relation to reaction centers, respectively dynamics. The authors might want to check the language.

[Reply]: We rephrase the sentence as “the breaking C–H bond is always directed away from (and uniformly distributed around) the center-of-mass of CH₄, which lies on the C atom.” (page 12, lines 17-19).

Page 12, Line 26

Shouldn't that be forward scattered CH₃ when the HF is scattered backward. Also see next sentence. I guess something got mixed up.

[Reply]: What you stated is a traditional way used in the community. However, this view will depend on which product (CH₃ or HF) is measured or referred to, and it could cause some confusion when different people talk about the dynamics from different views. For instant, stripping mechanism usually yields forward scattering, whereas rebound dynamics is associated with backward scattering. Then, seeing forward scattered CH₃ will argue a stripping mechanism, but the backward scattered HF will then suggest a rebound mechanism; an obvious dilemma. The situation could be more confusing when one is thinking of the angular distribution of concomitantly formed product state-pair (v_{CH_3} , v_{HF}), as the present case. This apparent paradox can be reconciled if one always views forward-scattering as the direction of the scattered product with respect to where it is from. For instance, CH₃ must come from CH₄ and HF from the F-atom (note that the directions of F-atom and CH₄ are opposite to each other in the center-of-mass frame), so that the forward-scattering will always convey the same dynamical behavior (say, stripping mechanism) irrespective of HF or CH₃ products.

Reviewer #2 (Remarks to the Author):

The authors (and others) have previously carried out extensive studies on the dynamics of the reaction $\text{F} + \text{CH}_4 \rightarrow \text{CH}_3 + \text{HF}$ and deuterated analogs, in which they have probed CH₃ products in their vibrational ground state and in various umbrella-mode (v_2) vibrationally excited states ($v_2 = 0, 1, 2, 3$) via REMPI and velocity-map imaging. These studies have revealed the pair-correlated scattering distributions of the CH₃ with various vibrational states of the HF (or HD) coproduct. The present study extends this work by using VUV photoionization at 118 nm as a universal probe to enable simultaneous detection of all rovibrational states of CH₃ formed in the chemical reaction. The new experimental results are interpreted with the help of comprehensive six-dimensional quantum dynamics simulations.

In addition to the product channels previously characterised, the paper identifies, and importantly, quantifies contributions from asymmetric-stretch (ν_1) excited CH_3 products. These have been observed before using REMPI detection, but one-photon ionisation makes it much more straightforward to determine the relative contributions from different product channels. The authors present product state distributions and pair-correlated angular distributions in excellent agreement with the theoretical predictions, and provide a detailed commentary on the mechanisms leading to the observed angular scattering.

This is a comprehensive, generally well-written, and very high quality piece of work, which fills in some additional dynamical details for the benchmark $\text{F} + \text{CH}_4$ reaction. However, I was extremely surprised at the number of times the authors stated the novelty of their 118 nm universal ionization approach, given that this ionization method, as well as universal ionization using other VUV wavelengths, has been used in velocity-map imaging studies in a number of different laboratories for well over a decade. A few examples of which I am immediately aware include Fan et al, J. Chem. Phys., 123, 204301 (2005), Ashfold et al, J. Phys. Chem. Lett, 8, 3440 (2017), Gardiner et al, Phys. Chem. Chem. Phys., 17, 4096 (2015), Lipciuc et al, J. Chem. Phys., 147, 013941 (2017), and various other papers from these and other groups, but I am sure there are numerous others. None of this previous work was mentioned or referenced in the manuscript. In my view, this is the one major shortcoming of the paper as it stands. There are also a few minor typos, etc, that the authors may want to correct in a revised version of the manuscript.

[Reply]: Yes, VUV one photoionization method is a well-established soft/universal detection scheme. We are fully aware of huge amounts of literatures. In fact, it has been mentioned on page 3, line 17, and some relevant prior works on crossed-beam scattering were cited as refs. 8-10.

Our message is more on the non-resonant one-photon ionization probe, rather than on a specific wavelength, say 118 nm. We chose 118 nm photon to probe CH_3 mainly because of its feature of potentially state-independent ionization efficiency, which will be one of the prime considerations for future application to any other system. Of course, easy generation of 118 nm also helps; many other VUV lasers at different wavelengths can also serve as probe but will be more involved to generate them.

The suggested publications concern instead photodissociation studies using 118 nm, which admittedly are mostly missing here not because of their unimportance, but rather considering the focus of this report (crossed-beam reactions). Nonetheless, we did discuss the photoionization detection of CH_3 in length about its sensitivity and quoted refs. 20-22. In that regard, we now gratefully add the suggested work by Gardiner et al in 2015 as ref. 23, which has unfortunately escaped our attention in the initial submission. We appreciate the reviewer for calling our attention to it.

1. Abstract: remove the initial ‘The’. Also, “electron-impact ionization” is now usually referred to simply as “electron ionization”.

[Reply]: Thanks, reword them accordingly.

2. Page 3, line 13: “a state-resolved detection” should just be “state-resolved detection”.

[Reply]: Thanks, correct it accordingly.

3. Page 11, line 8: lines 8-11 could do with a small amount of rewording, for clarity, e.g. “Thereby, the discrepancies in the forward peaks, the imprint of the resonance pathway, would be reduced if the reactants $\text{CH}_4(j = 1,2)$ were also included in the calculation. More significantly, the overall

distribution evolves from being predominantly forward peaking for (0₀,3) to exhibiting marked sideways scattering for (2₂,2), and then to essentially backward-scattering for (0₀,2)."

[Reply]: Thanks, correct it accordingly.

4. Page 11, line 16: "recoiled-energy" should be "recoil energy"

[Reply]: Thanks, correct it accordingly.

5. Page 12, line 9: I am not sure what the authors mean by "The CH₃-moiety does not directly invoke in bond cleavage and formation". Do they mean that the CH₃ is not directly *involved* in bond cleavage and formation?

[Reply]: Thanks, correct it accordingly.

6. Page 12, line 11: should be "...products attests to the strong coupling..."

[Reply]: Thanks, correct it accordingly.

7. Page 12, line 15: I am not sure what the authors mean when they say that the transferred H atom "is always sprouting away" from the centre of mass of CH₄. Are they just referring to the chemical structure of CH₄? If so, it might be clearer to say that the breaking C-H bond is always directed away from the centre of mass of CH₄, which lies on the C atom. However, I find it difficult to think of an example in which a breaking bond would not be pointing away from the centre of mass of the molecule within which it resides, so perhaps the authors could find a clearer way of making their point in this regard, perhaps with the aid of a diagram?

[Reply]: Thanks, see also the comments by Reviewer #1. We now rephrase the sentence as "the breaking C–H bond is always directed away from (and uniformly distributed around) the center-of-mass of CH₄, which lies on the C atom." (page 12, lines 17-19).

Reviewer #3 (Remarks to the Author):

The VUV single photon ionization method has been well known as a soft and universal product detection scheme, while universal detection has been linked with no product state specificity all the time previously. In this study, Pan et al. demonstrated an experiment combining the universality of VUV photoionization for the reaction product detection and the high velocity resolution achievable in a three-dimensional VMI which can handle relatively large reaction or ionization volumes of the reaction products, and they achieved universality and state-specificity for product detection simultaneously. They successfully mapped out almost all the important product pair-correlated dynamical attributes of the prototypical polyatomic reaction $F + CH_4 \rightarrow CH_3(vi) + HF(v)$ from a single product image. Six-dimensionality quantum dynamics calculations are performed and found to be able to reproduce nearly all the aspects observed in the experiment, and have provided new insights into this important prototypical polyatomic reaction. Such a method is a very useful addition to the community of reaction dynamics, and can be definitely used in the studies of many other reactions, I thus recommend its publication in Nature Communications. The whole paper is very nicely presented, and I only have two comments for the authors to consider.

[Reply]: Thanks for the positive comments. Our messages are clearly conveyed.

1. The universal VUV single photoionization detection is an important addition to the product state-specific detection method, and can provide most of the time qualitative information for the full

dynamical picture. However, its ability may not be overstated. In the study of the $F + CH_4 \rightarrow CH_3(v_i) + HF(v)$, one of the reasons why it is very successful is that the products are all rotationally cold, which enables the resolve of the pair-correlated vibrational distributions. For reaction products with significant rotational excitation, the complete resolve will become difficult. At the same time, previous state-specific studies also play an important role in analyzing the data in this study. I thus believe that the proper combination of the universal and state-specific methods would become a powerful method in studying reaction dynamics in the future, as demonstrated in this study.

[Reply]: We fully agreed. No detection scheme is perfect in all aspects for all reactions, and every probe scheme will end up with some compromises, depending on what one would like to achieve. We believe the $F+CH_4$ reaction is not the only one suitable for the present approach. We have several other classes of reactions in mind. Our intention is instead to alert the community about the possibility that a universal probe can also work as state-specific detection for some reactions by fully utilizing the 3D-VMI detection.

As stated in the end of conclusions (page 14, lines 18-21), attentive readers may also realize that the combination of the present non-resonant one-photon ionization and widely used REMPI scheme should really be the way to go. One may first do one-photon ionization VMI. Once the big picture has been sketched, one can always zoom in the details if necessary or when the required high-resolution REMPI information becomes available.

2. For the last paragraph on page 12, a classical physical picture is described to explain the “observed correlation that low umbrella-excited CH_3 prefers backward scattering, whereas higher umbrella-excited CH_3 yields more in the sideways and forward directions.” Two points that may need be explained a little more: first, it seems that both the large and small impact collisions involve the flatten motion of the CH_3 moiety, while one results in the umbrella vibrational excitation, and the other dose not? Second, the above picture only applies to processes within the same HF vibrational excitation, for example, for $HF=3$, the products are mainly forward scattered, even though the umbrella motion of CH_3 is not excited. Is it possible to make it clearer why this picture strongly depends on the excitation of the co-product HF ?

[Reply]: First, as presented in the paragraph starting from line 22 of page 12, the difference between the large and small impact parameter collisions is the magnitudes of the line-of-center force, which distorts the pyramidal geometry of CH_3 moiety to different extent as the C-H is ruptured to form HF . Large impact parameter collision exerts little distortion of the transition state structure of CH_3 moiety. As the newly formed HF departs, the abrupt change of the umbrella angle from pyramidal to planar geometry yields v_2 -excited CH_3 coproducts. For small impact parameter collision, the incoming F-atom pushes the attached H-atom and the central C-atom moves inward, gradually flattening the CH_3 moiety. As HF bounces backward, the relaxation of gradually flattening CH_3 moiety yields less v_2 -excitation.

Second, as explained in the second paragraph starting from line 19 on page 11, the HF vibrational angular distribution is mainly governed by the angle-angular momentum correlation. Qualitatively, $F+CH_4$ can be approximated as a simpler three-atom system if only the newly-formed HF product is of concern. On the other hand, unlike $A + BC$ reaction, the structure of the CH_3 moiety at the transition state and how it changes to product will also be at work, which can be comprehended by the line-of-centers argument in the paragraph starting from line 22 on page 12, as well as the elaborations given above. Although the above two arguments underpinning the dynamical behaviors for $HF(v)$ and $CH_3(v_2)$ cannot be completely decoupled, we believe that the formation of HF bond will be prime driving force and this early-barrier transition state with elongated H-F bond (line 20 on page 12) will always favor highly vibrational-excited HF products scattered in the forward direction, as evidenced

by Supplementary Fig. 5: HF($v=3$) is predominantly forward scattered and HF($v=1$) is mostly backward scattered, only HF($v=2$) exhibits angular distribution spread over a wide range of angles for deciphering the coincidentally formed CH₃(v_2) components. To emphasize the importance of the entire transition state structure, we rephrase the sentences starting from line 16 to 21 on page 13. Of course, the collision energy of this work is too low to produce much concomitantly umbrella-excited CH₃ with HF($v=3$). It will be interesting to do similar measurements at much higher collision energy to see if enough (v_2 (CH₃), $v_{\text{HF}}=3$) or (v_2 (CH₃), $v_{\text{HF}}=1$) could be formed and analyzed.

Reviewer #4 (Remarks to the Author):

The authors describe, implement, and test a VMI experiment capable of what they call “universal” state-resolved product selection. The well-studied F + CH₄ system is considered, and the experiment is shown to resolve product states of CH₃ alongside HF. The instrument is validated against quantum dynamics, which on its own is a substantial achievement of the work.

The demonstration given here is notable and, in my opinion, is appropriate for publication in a high-impact journal. Unfortunately, some awkward English and rushed descriptions make this short paper difficult to understand in places. I would specifically suggest improving the English in the abstract and elaborating on key discussion points. As written, the reader needs to already be familiar with VMI studies of this system to understand the paper.

[Reply]: Thanks for the positive comments about this work.

In the revision we tried to improve our writings in various places including the abstract, and elaborated some statements. We hope the revision is more readable to the general readers.

For example, the notation of the observed state pairs, e.g., (23, 2), are not explicitly explained and instead must be inferred.

[Reply]: It is now explained in the revision, page 6, line 17 (in the caption for Fig. 1 C).

I would suggest clarifying what aspects of the new design are responsible for the breakthrough here. The earlier work did not get information for all the vibrational levels of CH₃ at once, like here, but I wasn't sure why. Likewise, is there additional information in this experiment, as compared to earlier “3D” VMI studies on this system, or is the advance here that the information is gathered more directly?

[Reply]: The key of this work is to fully utilize the space-focusing capability as elaborated on page 4, line 29 to page 5 line 4. See also the above reply to Reviewer #1: “In retrospect, the potential of the present approach to achieving both universal and state-specific detection of crossed-beam scattering products is so obvious that we were surprising that no one, including ourselves, has made serious attempts over past years.”

All previous works used REMPI to probe CH₃ vibrational states, one at a time. Because of the disparate REMPI sensitivities, it is difficult to piece that information together to get a full dynamics picture. The present approach probes all CH₃ states simultaneously with nearly uniform detection sensitivities, and thus can uncover previously inaccessible information. For example, product vibrational branching fractions (which requires to know the relative sensitivities when detecting different vibrational states) reveal that the ground-state CH₃ products account for, unexpectedly, mere 40% of reactivity—despite being the most prominent feature in REMPI spectra. Most reactive fluxes are instead channeled into umbrella-excited CH₃ with little or negligible amounts for three other

vibrational modes, etc. The ability of both universal and state-specific detection also allows us to unveil the angle-angular momentum correlation in the reaction, and extend to whole class of $X+CH_4$ reactions.

Finally, I wondered if plots like Fig. 1A could be produced from the calculations?

[Reply]: We are not sure such a plot will add more information. All dynamical attributes extractable from Fig. 1A are already presented in Figs. 2-4 and compared with corresponding theoretical calculations in a quantitative manner. Such a plot from theory seems redundant and will be a qualitative comparison at best (besides, it needs to be convoluted first by density-to-flux correction for a direct comparison with Fig. 1A).

Reviewer #1 (Remarks to the Author):

The authors answered my questions and cleared up some confusion. The changes to the text make the manuscript easier to understand. The manuscript is suitable for Nature Communications. Maybe, the authors want to consider some minor comments:

[Reply] Thanks for the positive feedback.

In the abstract and on page 5 they refer to “previously inaccessible insights”. Why not state at this point what was previously inaccessible more plainly. The wording works in the abstract but on page five it reads strange with the kind of left confused at as to what is new- The answer is only given later on page6.

[Reply] We found that it will be too involved to explain “what is new” at that point. That is the reason for the original statement on Page 5, Line 27: “As will be presented (Note: change now from “shown”) below, the results from such universal and state-specific detection in turn give access to hitherto inaccessible information.”

I now understand the author's use of forward and backward in the relative frame of reference. Maybe readers would benefit from an addition of the relative orientation of the incident beams with respect to the scattering angle as indicated in figure1.

[Reply] We are delighted that we are in line with each other. As suggested, Fig. 1 is modified.

I find “all-important” a presumptuous phrase to use. Why use it at all? For me, it adds nothing to the paper but confusion because it has no direct scientific meaning.

[Reply] Two most important information can be derived from the chemical dynamics studies are the product state and angular distributions. Traditionally, these two attributes in reactions beyond the simple $A + BC$ system are, unfortunately, acquired by very different and usually non-compatible detection schemes. The approach proposed in this work enables simultaneously acquiring both dynamical attributes in a single measurement, and thus providing “all-important” information. We originally used “all-important” simply as a plain statement to reflect this fact, and when checking the dictionary, this phrase also does not sound presumptuous. Nonetheless, we now change “all-important” to “all essential”, and we welcome any suggestion for a better phrase to use here.

A comment:

In the authors refer to space and length of the communication with respect to the limited number of references to earlier studies. Length might be an argument, number of allowed references is not. The author guidelines state that up to 70 references are allowed or considered reasonable. The current number of references is 44 (three more than in the

original submission). I still think this is a major point to consider. The authors do not need to write a “review” style introduction but the references to prior studies on either the reaction studies or the methodology are limited.

[Reply] We now elaborate a bit about prior works on Page 5. Lines 17-21 (reproduced below), and some relevant new references 20-27 are cited.

“This reaction has been the subject of extensive studies over the past decades for its fundamental significance as a prototypical polyatomic reaction as well as its practical application in chemical lasers²⁰⁻²⁷. Dynamically most of prior works focused on the product state distributions of either CH₃ or HF (refs 23-26), except more recent studies on the angular distributions through REMPI-tagged product pair-correlation measurements¹³⁻¹⁶.”

Reviewer #2 (Remarks to the Author):

The authors have responded comprehensively to the suggestions of the referees, and amended their manuscript accordingly. Assuming the other referees are in agreement, I believe that the paper is now suitable for publication in Nature Communications.

[Reply] Thanks for the suggestions and comments which make the paper better.

Reviewer #3 (Remarks to the Author):

The authors have addressed all my comments in proper ways, I thus recommend its publication in Nature Communications in its current form.

[Reply] Thanks for the suggestions and comments which make the paper better.